Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/27/2013
PI Name:	Kozlowski, Steve Ph.D.		
Project Title:	Monitoring and Regulating Teamwork		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	33620	**Congressional District:**	12
Comments:	I moved from Michigan State University to the University of South Florida in August 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	09/05/2012	**End Date:**	09/04/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/4/14 per NSSC info (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Biswas, Subir (Michigan State University) Chang, Chu-Hsiang (Michigan State University)		
Grant/Contract No.:	NNX12AR15G		
Performance Goal No.:			
Performance Goal Text:			
	This continuation proposal is for ground-based research to address: PRD Risk of Performance Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team. IRP Gap - Team2: Given the context of long duration missions, what are the best tools to effectively monitor and measure task performance, teamwork, and psychosocial performance? Collaboration and cohesion are critical underpinnings of teamwork. This is especially true for high reliability teams such as long duration (LD) space flight crews that have to perform in extreme environments. Effective teamwork is essential		

Task Description:	for minimizing errors and supporting team performance, and it is reflective of good psychosocial adaptation to the stresses of LD space missions. Research Plan: Our research has two key goals: (a) benchmarking variation in team collaboration and team cohesion over LD missions and (b) developing a monitoring and regulation system to unobtrusively measure team collaboration and cohesion. For (a), there is substantial research that documents a relationship between team cohesion and effectiveness, however, that evidentiary foundation is based on static data. We are systematically examining variation in team cohesion and collaboration in LD analog teams operating in isolated, confined, and extreme (ICE) environments. We have collected preliminary data for ICE teams operating in the Antarctic and have detected evidence for considerable variation in cohesion levels over time. We seek to extend this research to larger samples and longer durations to help calibrate expected levels of cohesion variability, which is essential for developing appropriate monitoring models. For (b), one key component of the monitoring system is a technology - a wearable sensor network - that assesses the frequency, duration, distance and quality of collaboration as team members work together. We have functional prototypes and have initiated a series of basic lab validation studies. At the end of the current award, we will have developed preliminary validation evidence to demonstrate that the monitoring technology is capable of accurately capturing team member interactions (relative to video coding of interactions) in highly structured collaborations. Additional steps needed to develop the monitoring and regulation system include: (a) extending validation research of the monitoring technology to capture successively more complex and naturalistic team interaction dynamics, and (b) initiating the development of a feedback “dashboard” to display multiple data streams (collaboration frequency, duration, and distance; physiological arousal) that can diagnose the effectiveness of collaboration and teamwork. Specific Aims: The proposed work has been developed in consultation with the Element Senior Scientist responsible for current research and this proposed extension. The purpose of this proposal is to (1) extend our benchmarking of LD team functioning in ICE environments, (2) extend the validation process for the monitoring and regulation system and begin to develop a data fusion model, and (3) develop a feedback dashboard to display multimodal data on team functioning. Project deliverables include (1) benchmarking data for long duration analog teams in operating in ICE environments, (2) validation evidence to establish proof-of-concept for the monitoring technology and preliminary development of a data fusion model, and (3) software updates to provide a feedback display to diagnose team functioning based on multimodal monitoring data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Team cohesion is not just a critical factor for astronaut teams and ground crews; cohesion is important to the effectiveness of all teams and especially those that operate in critical, high reliability setting. Of the many team process factors that support team effectiveness, team cohesion is the most studied with over a half century of research. Yet, remarkably, very little is known about the characteristics that promote its development and maintenance. For example, we know that experience working together is associated with cohesion formation and maintenance, but what are the mechanisms? Teams that do not cohere replace problematic members or disintegrate so experience only reveals those teams that survive, but that does not tell us why or how. This research, which will uncover the dynamics of collaboration, cohesion, and effective team functioning and will create technologies to monitor team cohesion and guide interventions to restore it, has the potential for wide utility in aviation, military, medical, industrial, and other environments where society depends on the effective performance of high reliability teams.
Task Progress:	A substantial amount of cross-sectional (i.e., static) research has shown a strong relationship between team cohesion and team performance. Although team cohesion is generally assumed to be a relatively stable team process variable, little is known about its potential dynamics. Across different types of benchmarking data (military teams during 8 weeks of intensive training), simulation teams (across hours to weeks of performance), problem solving teams (hours), and analog teams (6 to 20 + weeks deployed in an ICE environment), we observed considerable variation in team cohesion and team collaboration effectiveness. This convergent evidence across very different teams, tasks, and timeframes makes it evident that the assumed stable relationship between team cohesion and team effectiveness needs to be examined closely and unpacked. This research is developing metrics, tools, and techniques to do just that.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
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